

Customer Insights Dashboard

End-to-End Data Analytics Report

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Tools: SQL, Python, Power BI (DAX)

Executive Summary

This project demonstrates an end-to-end analytics workflow using SQL for exploratory analysis, Python for data preparation, and Power BI for visualization. A dataset of 3900 customers generated total revenue of \$233,081. Business-ready KPIs were created using DAX measures and validated through SQL-based analysis.

SQL-Based Exploratory Data Analysis

SQL was used to validate trends, aggregate metrics, and uncover customer behavior patterns before building dashboards in Power BI. The following queries highlight key analytical steps performed directly on the dataset.

1. Net Revenue by Gender

This query calculates total revenue generated by each gender to understand purchasing power distribution across customer demographics.

```
SELECT gender, SUM(purchase_amount_usd) AS Net_Revenue FROM  
customer_shopping_behavior GROUP BY gender;
```

Insight: Male customers generate higher total revenue compared to female customers, indicating a stronger revenue contribution within this dataset.

2. Average Review Rating by Item Purchased

This query computes the average customer review rating for each product, helping identify top-performing and underperforming items based on customer satisfaction.

```
SELECT item_purchased, ROUND(AVG(review_rating), 2) AS Net_Rating FROM  
customer_shopping_behavior GROUP BY item_purchased ORDER BY Net_Rating DESC;
```

Insight: Products such as Gloves and Sandals receive the highest average ratings, suggesting better customer satisfaction for these items.

3. Subscription Status – Sales & Customer Distribution

This query compares subscribers and non-subscribers in terms of customer count, average sales value, and total revenue contribution.

```
SELECT subscription_status, ROUND(AVG(purchase_amount_usd), 2) AS Net_Sales,  
COUNT(DISTINCT customer_id) AS Number_of_Customers, SUM(purchase_amount_usd) AS  
Total_Revenue FROM customer_shopping_behavior GROUP BY subscription_status;
```

Insight: While non-subscribers contribute a larger share of total revenue due to volume, average sales per customer remain comparable between both groups, highlighting potential for subscription conversion strategies.

4. Top Items by Category Using Ranking

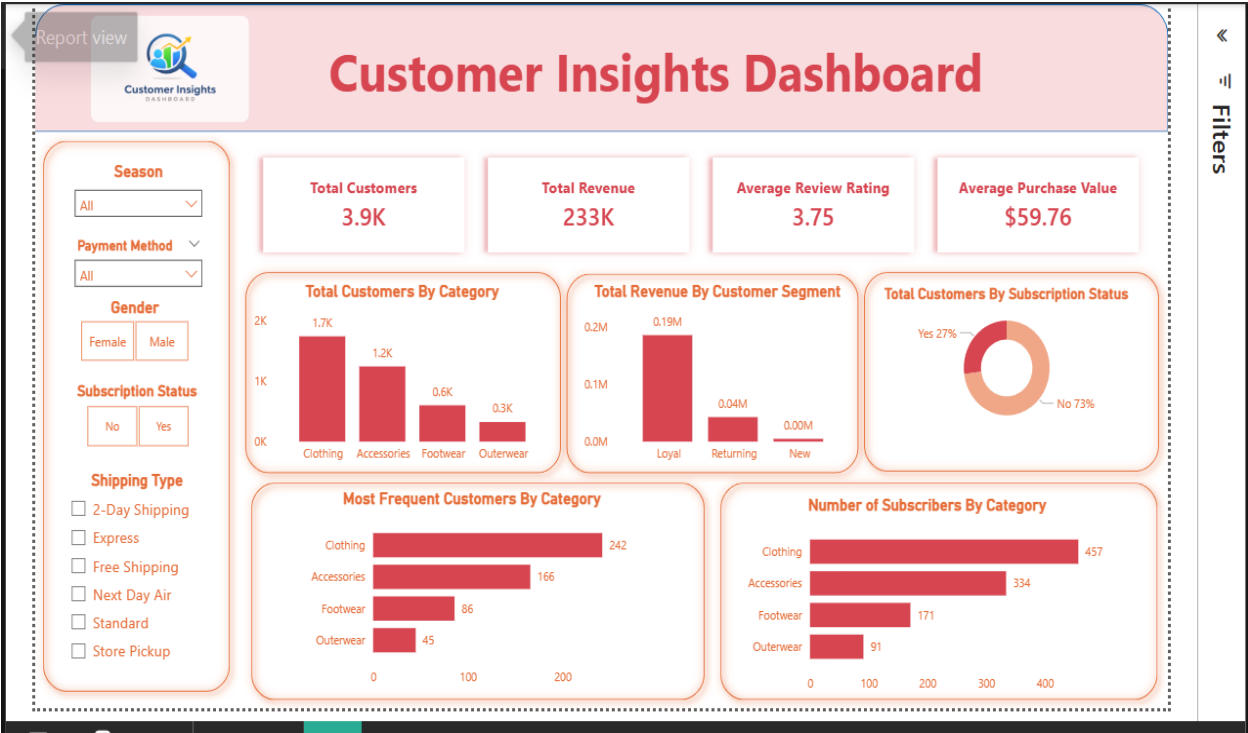
Window functions were used to rank products within each category based on purchase frequency, helping identify the most popular items per category.

```
SELECT item_purchased, category, COUNT(item_purchased) AS item_count, RANK() OVER  
(PARTITION BY category ORDER BY COUNT(item_purchased) DESC) AS item_rank FROM  
customer_shopping_behavior GROUP BY item_purchased, category;
```

Insight: Items such as Jewelry (Accessories), Blouse (Clothing), Sandals (Footwear), and Jacket (Outerwear) rank highest within their respective categories.

Dashboard Validation Using SQL Insights

The insights derived from SQL analysis directly informed the KPIs and visuals implemented in Power BI. SQL results were cross-validated with DAX measures to ensure accuracy and consistency across the analytical workflow.



Conclusion

This project showcases strong proficiency in SQL-driven analysis, KPI development using DAX, and interactive dashboard creation in Power BI. The combined use of SQL, Python, and Power BI demonstrates an industry-aligned, end-to-end data analytics workflow.